

Application Test Report No. 1

Date: 2026.06.14

A. Description of Activities Performed

Environments and devices used for testing:

- **System:** Windows 11, **Browser:** Google Chrome v149.0.7827.104 (Resolution: 1920x1080)
- **System:** Windows 11, **Browser:** Mozilla Firefox v151.0.3 (Resolution: 1920x1080)
- **Build number:** v2.0.1 **dated** 02.06.2026

Application areas and types of tests performed:

Application Area	Types of Tests Performed	Time Spent on Testing
User Interface	Functional tests (verification of operation at 1920x1080), unit tests	1 h
Session Trends Module	Functional tests (calculation of minimum, maximum, and average values; selection of time ranges), unit tests	1 h
Statistical Analyses	Functional tests (tests of correct calculation: median, mode, standard deviation, and coefficient of variation), unit tests	0,5 h
Fetching data from NBP API	Functional tests (API communication tests, error notification tests, dates before 2002-01-02)	1 h

B. Summary of All Defects

1. **Fixed and retested:** (not applicable)
2. **To do:**
 - a) Missing recent dates for analysis (the last available date is 02.06, which is the release date).
 - b) Missing request chunking for API queries.
 - c) Missing an appropriate error message in case of no internet connection.
3. **Risky to fix at the moment:** (none)

C. Detailed Description of Defects

Test 2a

- **Discovered by:** Aliaksandr Yurusau
- **Report in which discovered:** 1
- **Priority:** Highest
- **Repeatability:** Always
- **Browser:** Not applicable
- **Steps to reproduce:**
 1. Open the application in a browser.

2. Attempt to change the start date to 03.05.2026 for 1 month period.
- **Test result:** It is impossible to select a date and obtain results for dates after 02.06.2026.

Test 2b

- **Discovered by:** Aliaksandr Yurusau
- **Report in which discovered:** 1
- **Priority:** Medium
- **Repeatability:** Always
- **Browser:** Not applicable
- **Steps to reproduce (e.g., Google Chrome):**
 1. Open the application in a browser.
 2. Select any start date that is at least one year earlier than today's date.
 3. Select an analysis period of 1 year.
 4. Open the developer console by pressing F12.
 5. Go to the Network -> Fetch/XHR tab.
- **Test result:** A single request to the API covering the entire period can be observed (e.g., 2020-01-01/2021-01-01). This does not comply with **system constraint #13**: *"the system shall be constrained to fetching a maximum of 93 days of historical data per request. If a user selects a period longer than 93 days, the system shall perform request chunking."*

Test 2c

- **Discovered by:** Aliaksandr Yurusau
- **Report in which discovered:** 1
- **Priority:** Low
- **Repeatability:** Always
- **Browser:** Not applicable
- **Steps to reproduce (e.g., Google Chrome):**
 1. Open the application in a browser.
 2. Open the developer console by pressing F12.
 3. Go to the Network tab.
 4. In the throttling dropdown, select "offline" instead of "no throttling".
 5. Change the analysis data (e.g., change the period to 1 week).
- **Test result:** The message failed to fetch data: failed to fetch is displayed, which is a standard error from the code. A proper message must be provided instead (e.g., "Offline").

D. Summary of Exit Criteria for All Tests

1. **Did we manage to implement the planned tests?** Yes, all planned functional, acceptance, and NBP API communication tests were successfully completed. The application was tested in the target environment: Windows 11 system at 1920x1080 resolution using Chrome and Firefox browsers.
2. **Did any difficulties occur?** No issues preventing the execution of the tests were recorded. The environment itself (both local and the external NBP API) operated stably.
3. **Verification of work time estimates:** The time spent on testing aligns with the initial assumptions (approximately 4 hours in total). The majority of the time was spent on tests

verifying the API operation and edge cases regarding date ranges, which resulted in finding several defects.

4. **Realized risks and newly discovered risks:**

- *Realized risk:* Developers took shortcuts regarding the external API handling “the system constraint concerning request chunking (splitting queries into smaller packages) was ignored. This poses a risk of our application being blocked by NBP if the API tightens its limits in the future.
- *New risk:* Risk of outdated data (a "hardcoded" end date tied to the release creation date of 02.06.2026). If this is not fixed, the application will become completely useless after a few months because it will not allow fetching the latest currency exchange rates.

5. **Conclusions for the future:** Documentation must be more strictly enforced during coding to avoid bypassing more challenging features such as request chunking.